

DAY-4

Tasks

1.

List 3 key differences between Ethernet cables and fiber optic cables, and give one real-world example where each type would be used (for example, home WiFi router, data center, college campus backbo

Difference	Ethernet Cable	Fiber-Optic Cable
Transmission	Uses electrical signals through copper wires.	Uses light signals through glass/plastic fibers.
Distance	Generally suitable for shorter network connections; standard twisted-pair Ethernet links are commonly up to about 100 m.	Suitable for much longer distances, especially in backbone networks.
Speed & interference	Offers high speeds but copper can be affected by electromagnetic interference.	Supports very high speeds and is resistant to electromagnetic interference.
Cost & installation	Usually cheaper and easier to install.	Usually more expensive and requires specialized equipment for termination.

2.

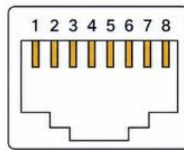
Draw and label a diagram showing the internal wiring of a straight-through Ethernet cable (use T568B wiring standard), indicating the color of each wire and its pin number.

RJ45 Wiring Diagram: T568A vs T568B

ETHERNET CABLE COLOR CODES



RJ45 CONNECTOR
8 POSITION / 8 CONTACTS



T568A		
1		White/Green
2		Green
3		White/Orange
4		Blue
5		White/Blue
6		Orange
7		White/Brown
8		Brown

T568B		
1		White/Orange
2		Orange
3		White/Green
4		Blue
5		White/Blue
6		Green
7		White/Brown
8		Brown



Use the same standard on both ends for straight-through Ethernet cables.

3.

Watch a YouTube video demonstrating how to use an RJ45 crimper, then write a step-by-step guide (in your own words) on how to attach an RJ45 connector to a CAT6 cable.

Step-by-step procedure

1. **Cut the CAT6 cable** to the required length.
2. **Remove the outer jacket** from the end of the cable using a cable stripper. Be careful not to damage the small internal wires.
3. **Separate the four twisted pairs** inside the cable.
4. **Untwist and straighten the eight wires** so they can be arranged properly.
5. Arrange the wires according to **T568B**:
 - White/Orange
 - Orange
 - White/Green
 - Blue
 - White/Blue
 - Green
 - White/Brown
 - Brown
6. **Keep the wires flat and in the correct order.**
7. Trim the wire ends evenly so that they can reach the front of the RJ45 connector.

8. **Insert the wires into the RJ45 plug**, making sure each wire enters its correct channel and reaches the end of the connector.
9. Check the order again before crimping. This is the moment where fixing a mistake is easy; after crimping, it's annoying.
10. Put the RJ45 connector into the **RJ45 crimping tool**.
11. Squeeze the crimper firmly so the metal contacts inside the connector pierce the individual wires and the cable jacket is held securely.
12. Finally, use a **cable tester** if available to check that all eight pins are connected correctly.

4.

Imagine you are setting up a gaming café with 10 PCs for LAN gaming. Decide which cable type (Ethernet or fiber optics) and which tool (RJ45 crimper or fiber cleaver) you would use, and explain your choices in 3-4 sentences.

Hint: Consider speed, distance, and budget for your answer.

Steps

1. Get 10 CAT6 Ethernet cables of suitable length.
2. If the cables don't already have RJ45 connectors, use an RJ45 crimper to attach an RJ45 connector to both ends of each CAT6 cable.
3. Arrange the wires using the T568B standard:
 - White/Orange
 - Orange
 - White/Green
 - Blue
 - White/Blue
 - Green
 - White/Brown
 - Brown
4. Connect one end of each Ethernet cable to the Ethernet port of each gaming PC.
5. Connect the other end of each cable to an Ethernet port on the network switch.
6. Connect the switch to the router using another Ethernet cable.
7. Connect the router to the ISP/modem/ONT so the café has Internet access.
8. Turn on all 10 PCs and check that they receive IP addresses from the router.

9. Test the network by checking Internet access and, for LAN gaming, confirming that the PCs can communicate with each other.
10. Use an Ethernet cable tester to check any newly crimped cables if a PC doesn't connect.

Final setup

ISP

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MODEM/ONT

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ROUTER

|



SWITCH

|— CAT6 — PC 1

|— CAT6 — PC 2

|— CAT6 — PC 3

|— CAT6 — PC 4

|— CAT6 — PC 5

|— CAT6 — PC 6

|— CAT6 — PC 7

|— CAT6 — PC 8

|— CAT6 — PC 9

|— CAT6 — PC 10